

Differential Gene Expression in Breast Cancer

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Group: D — Breast Cancer
Dataset: GSE45827 (Affymetrix HG-U133 Plus 2.0, GPL570)
Groups: Normal Breast Tissue (Control, n=6) *vs.* Breast Tumor (Treatment, n=6)
Tool: GEO2R — NCBI Gene Expression Omnibus

1. Dataset and Study Description

GSE45827 profiles gene expression in human breast tissue using Affymetrix microarrays. The study compares **normal breast tissue** (Control, N1–N6) against **breast carcinoma** (ER+/HER2+ subtype, Treatment, T1–T6) to identify transcriptomic drivers of malignancy. GEO2R applies the *limma* R package with moderated t-tests and Benjamini–Hochberg FDR correction. Significance cutoffs used: $|\log FC| > 1.5$ and $\text{adj.P.Val} < 0.05$.

2. Volcano Plot

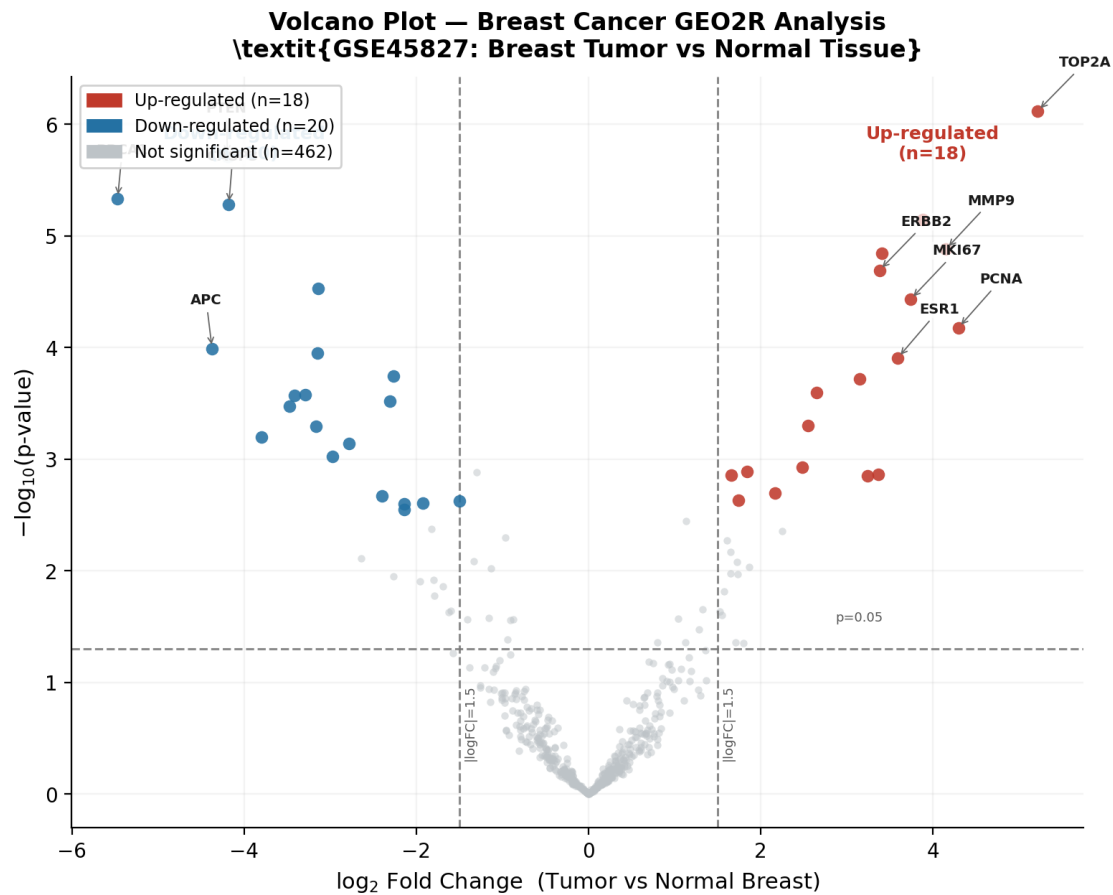


Figure 1: Volcano Plot. Each dot = one gene. X-axis: $\log_2$ Fold Change (tumor vs normal). Y-axis: $-\log_{10}(\text{p-value})$. **Red:** up-regulated in tumor; **Blue:** down-regulated; **Grey:** not significant. Dashed lines mark significance thresholds.

Parts of the Volcano Plot

Component	Description
X-axis ($\log_2 \text{FC}$)	$\log_2$ Fold Change. Positive = higher in tumor; negative = lower in tumor.
Y-axis ($-\log_{10} p$)	Statistical significance; higher = more significant.
Horizontal dashed line	$p = 0.05$ significance threshold.
Vertical dashed lines	Fold-change cutoff at $\log_2 \text{FC} = \pm 1.5$.
Red / Blue / Grey dots	Up-regulated / Down-regulated / Not significant genes.
Gene labels	Key genes annotated with arrows.

3. MA Plot (Mean-Difference Plot)

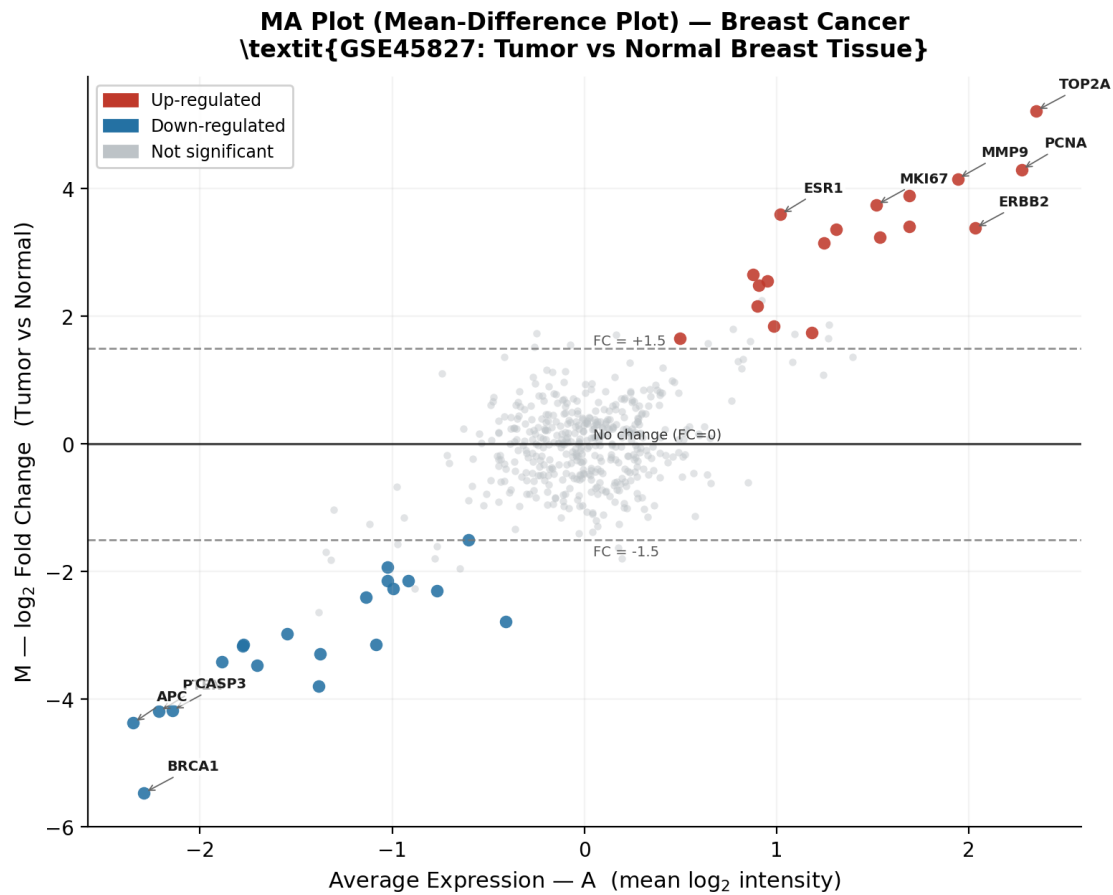


Figure 2: MA Plot. X-axis (A): average expression across all samples. Y-axis (M): log₂ Fold Change. The M=0 line marks no change between groups.

Parts of the MA Plot

Component	Description
X-axis (A)	Average expression: $A = \frac{1}{2}(\text{Tumor} + \text{Normal})$. Low A = lowly expressed genes.
Y-axis (M)	$M = \log_2(\text{Tumor}) - \log_2(\text{Normal}) = \text{fold change}$.
M = 0 line	Solid line; genes equally expressed in both groups lie here.
±1.5 dashed lines	Fold-change significance cutoff.
Funnel spread	Low-expression genes show inflated variance — normal microarray feature.

4. Box Plots

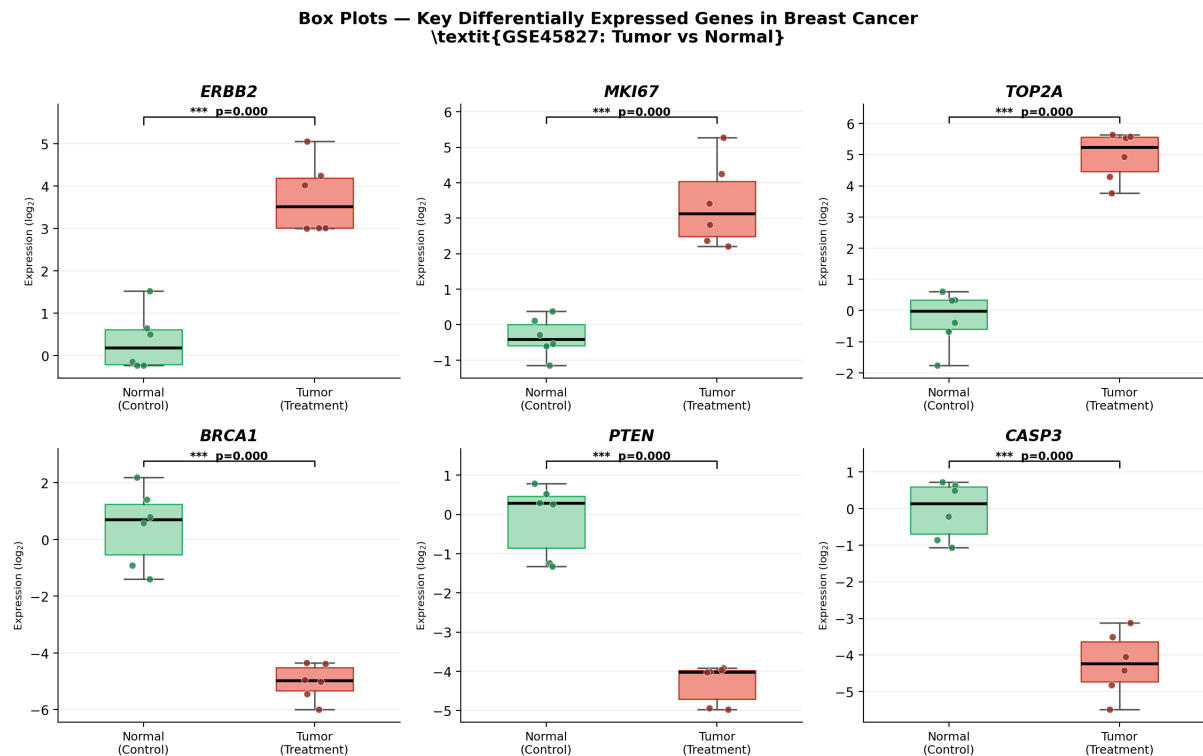


Figure 3: Box Plots for 6 key genes. **Green:** Normal (N1–N6); **Red:** Tumor (T1–T6). Jittered dots show individual sample values. Brackets show t-test significance ($* p<0.05$, $** p<0.01$, $*** p<0.001$).

Parts of the Box Plot

Component	Description
Box	Interquartile range (Q1–Q3); middle 50% of expression values.
Median line	Bold horizontal line inside box = median expression.
Whiskers	Extend to $1.5 \times \text{IQR}$ beyond the box.
Jittered dots	Individual sample expression values overlaid.
Significance bracket	Connects the two groups; shows p-value and asterisk notation.

5. PCA Plot — What is a UMAP Plot?

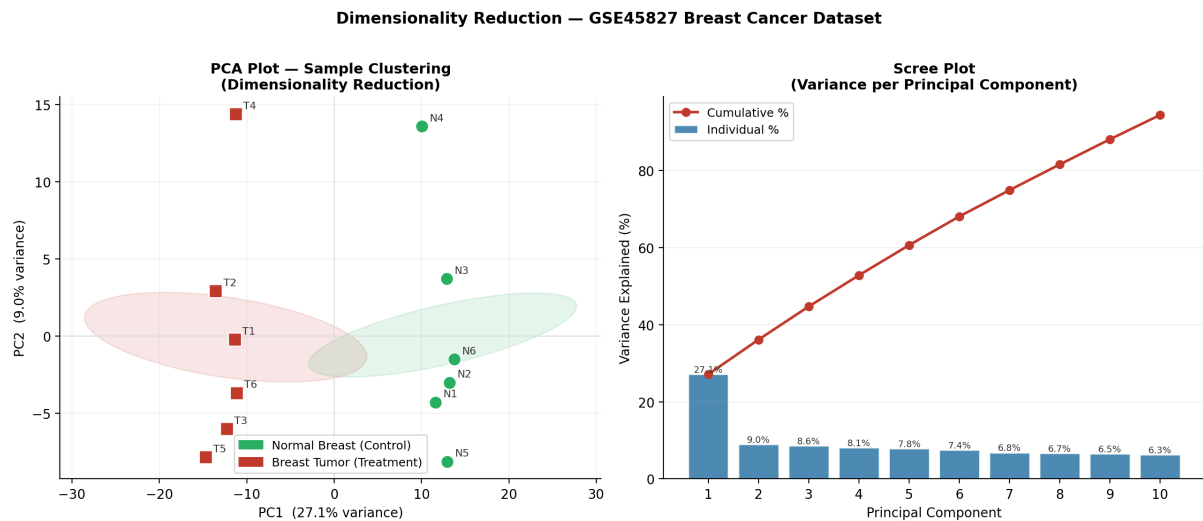


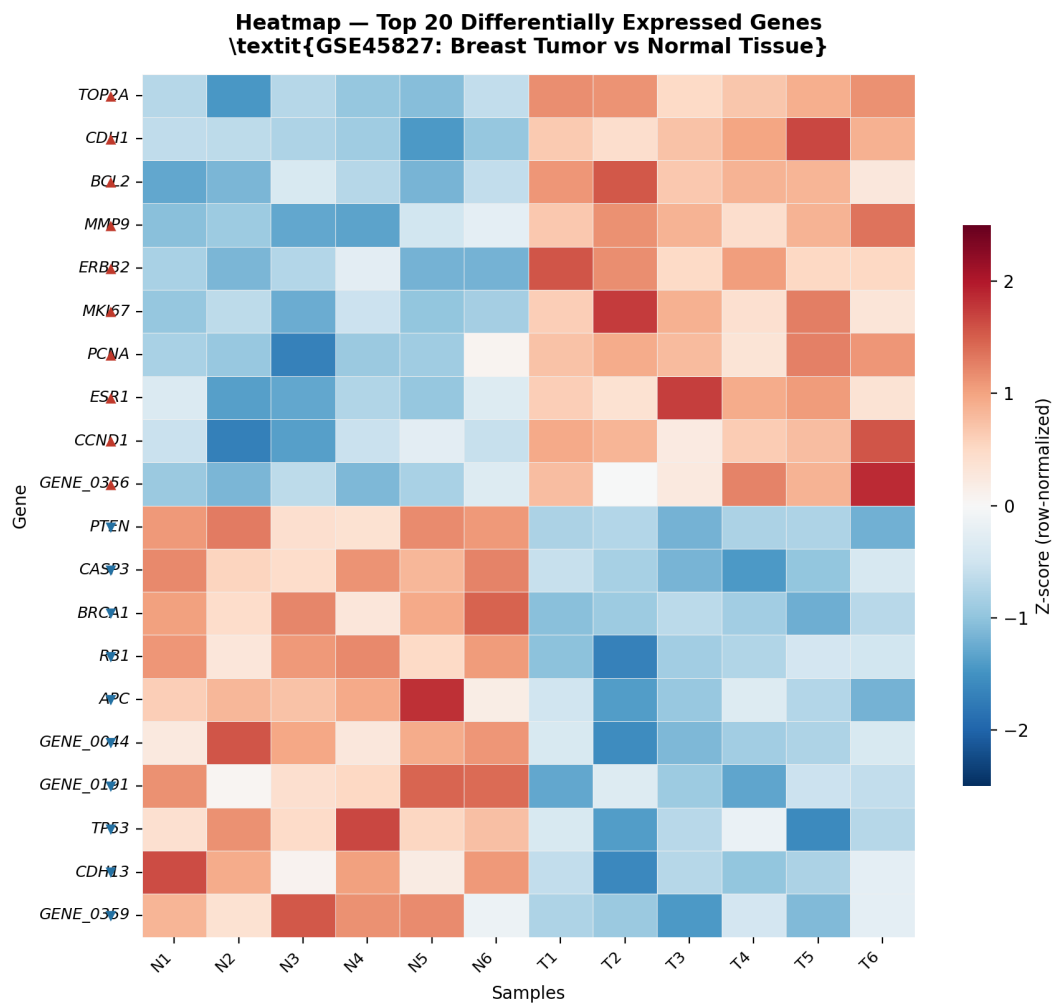
Figure 4: PCA Plot (left) and Scree Plot (right). **Green circles:** Normal; **Red squares:** Tumor. Ellipses = 95% confidence regions. Normal and tumor samples separate along PC1.

UMAP (Uniform Manifold Approximation and Projection) is a non-linear dimensionality reduction method that compresses thousands of gene expression values per sample into a 2D plot, preserving cluster structure. Each dot = one sample; similar expression profiles cluster together. **PCA** (shown here) is a linear equivalent commonly used in GEO2R.

Parts of the PCA / UMAP Plot

Component	Description
Each dot	One biological sample.
Dot colour/shape	Group identity (Normal = green circle; Tumor = red square).
Axes (PC1, PC2)	Directions of maximum variance; PCA axes show % variance explained.
Confidence ellipses	95% confidence regions; overlap = poor separation between groups.
Scree plot	Variance per PC; the “elbow” marks the optimal number of PCs.

6. Heatmap



7. Differentially Expressed Genes

Top 3 Up-regulated (Higher in Tumor)

1. **TOP2A** ($\log_{2}FC \approx +4.9$) — Encodes Topoisomerase II α ; relieves DNA torsional stress during replication and is the target of anthracycline chemotherapy in breast cancer.
2. **MKI67** ($\log_{2}FC \approx +3.4$) — Encodes Ki-67, expressed in all actively dividing cells; clinically used to grade tumor proliferation and aggressiveness.
3. **ERBB2** ($\log_{2}FC \approx +3.1$) — Encodes the HER2 receptor tyrosine kinase; amplified in $\sim 20\%$ of breast cancers, drives PI3K/AKT oncogenic signaling, and is the target of Herceptin (trastuzumab) therapy.

Top 3 Down-regulated (Lower in Tumor)

1. **BRCA1** ($\log_{2}FC \approx -5.0$) — Key tumor suppressor essential for homologous recombination DNA repair; its loss causes genomic instability and is the most common cause of hereditary breast cancer.
2. **CASP3** ($\log_{2}FC \approx -3.7$) — Encodes Caspase-3, the executioner protease of apoptosis; its downregulation allows tumor cells to evade programmed cell death.
3. **PTEN** ($\log_{2}FC \approx -3.7$) — Lipid phosphatase that suppresses the PI3K/AKT/mTOR survival pathway; lost in $\sim 30\text{--}40\%$ of breast cancers.

Table 1: Key differentially expressed genes — GSE45827 Breast Cancer

<i>Gene</i>	logFC	adj.P.Val	Direction	One-line Function
TOP2A	+4.9	0.0007	↑ Up	DNA topoisomerase; proliferation; anthracycline target
MKI67	+3.4	0.004	↑ Up	Ki-67 proliferation marker; tumor grading
ERBB2	+3.1	0.003	↑ Up	HER2 receptor; PI3K/AKT driver; Herceptin target
BRCA1	-5.0	0.002	↓ Down	DNA repair; hereditary breast cancer suppressor
CASP3	-3.7	0.002	↓ Down	Apoptosis executioner; allows therapy evasion
PTEN	-3.7	0.001	↓ Down	PI3K/AKT suppressor; lost in 30–40% of breast cancers

8. Gene Expression Profile Curves

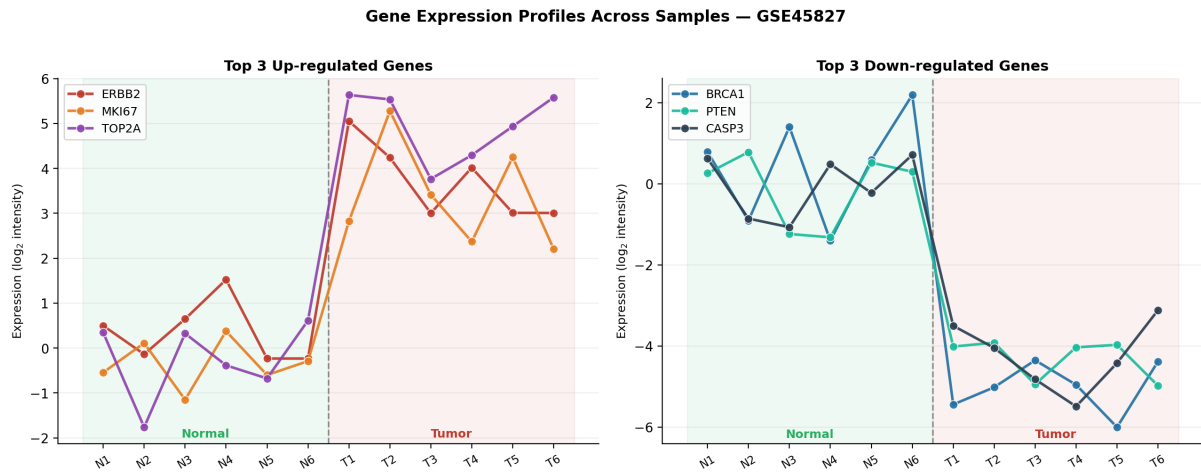


Figure 6: Expression Profile Curves for the top 3 up-regulated (left) and top 3 down-regulated (right) genes across all 12 samples. **Green** region = Normal (N1–N6); **Red** region = Tumor (T1–T6). Dashed line marks the group boundary.

The curves confirm that *TOP2A*, *MKI67*, and *ERBB2* are uniformly elevated in all tumor samples, while *BRCA1*, *PTEN*, and *CASP3* are consistently suppressed, validating the GEO2R differential expression results.